AI FOR GOOD · EDUCATION · ACCESSIBILITY

Phloem

AI Learning Companion for learners who understand by listening and speaking.

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Project: Phloem
Maker: Aaryan Kandiah
Demo: https://db0697e89cd74ce696d44f72edfcc11d.ap-singapore.myide.io/
Core AI: Z.ai GLM-5.1 · OpenAI vision & STT · Google/OpenAI/browser TTS

THE PROBLEM

Many learners understand spoken explanations,

but get blocked by written materials.

01
ESL & EFL

Learners studying in a non-native language

02
Dyslexia

Students who face reading difficulty

03
Young & Low-Literacy

Learners who need patient guidance

04
Solo Studiers

People studying alone without a tutor beside them

Existing tools translate or answer. *They rarely facilitate learning.*

OUR INSIGHT

Use the camera, ears, and voice — not reading fluency.

1. Show or upload a learning page
2. AI extracts topic, text & diagram context
3. Tutor gives spoken Socratic guidance
4. Learner answers aloud
5. Phloem scores a retrieval check & updates progress

***Phloem** is a voice-first tutor that reads a page, teaches it through guided questions, and tracks whether the learner actually understood.*

One learner. One page. One spoken question.

01 Start on Voice Learning

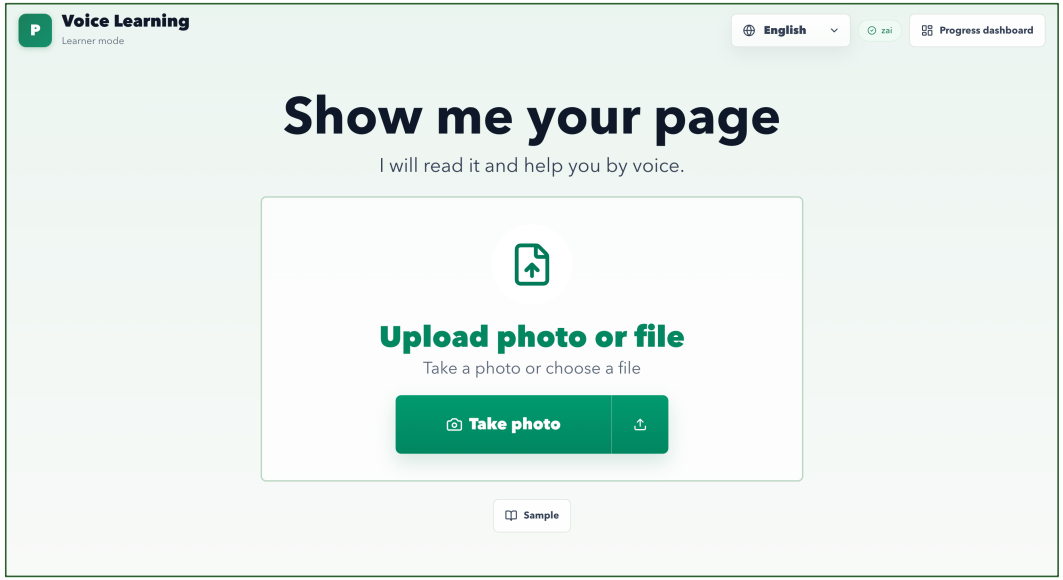
02 Click Upload Photo or File

03 "What do plants need for photosynthesis?"

04 Show spoken tutor answer + speech trace

05 Answer follow-up: "Carbon dioxide"

06 Open Progress — concept update appears

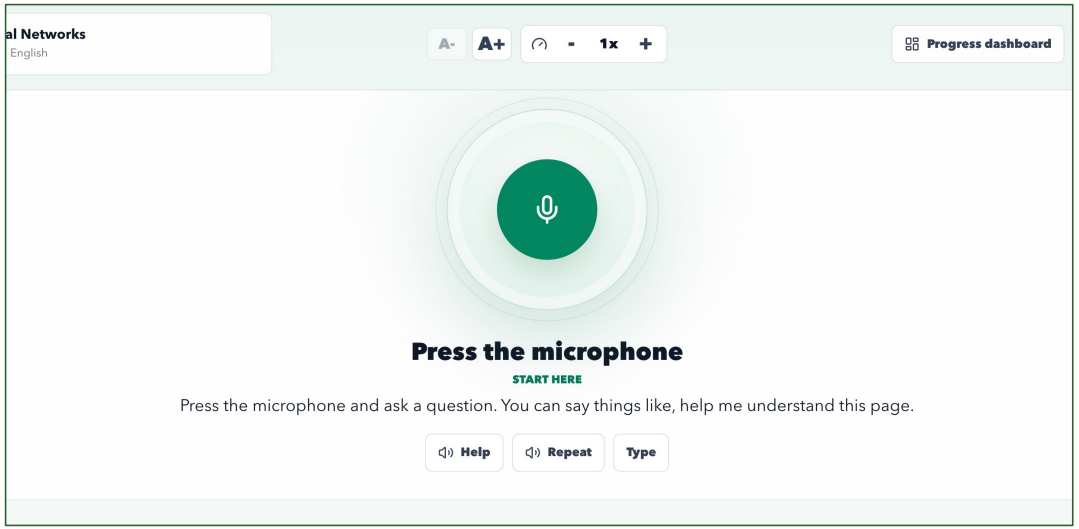


One learner, one page, one question, one progress update.

PHLOEM

HUMAN-CENTERED DESIGN

Designed around listening — not reading.



- **One target**
  Single centered upload or photo capture
- **One mic**
  Central microphone affordance — nothing to hunt for
- **Minimal actions**
  Help, Repeat, Type, Skip
- **Spoken-friendly**
  Large, readable tutor text
- **Learner controls**
  Tutor voice speed and text size
- **For caregivers**
  Progress dashboard available, never in the way

VOICE LEARNING + FULL DASHBOARD

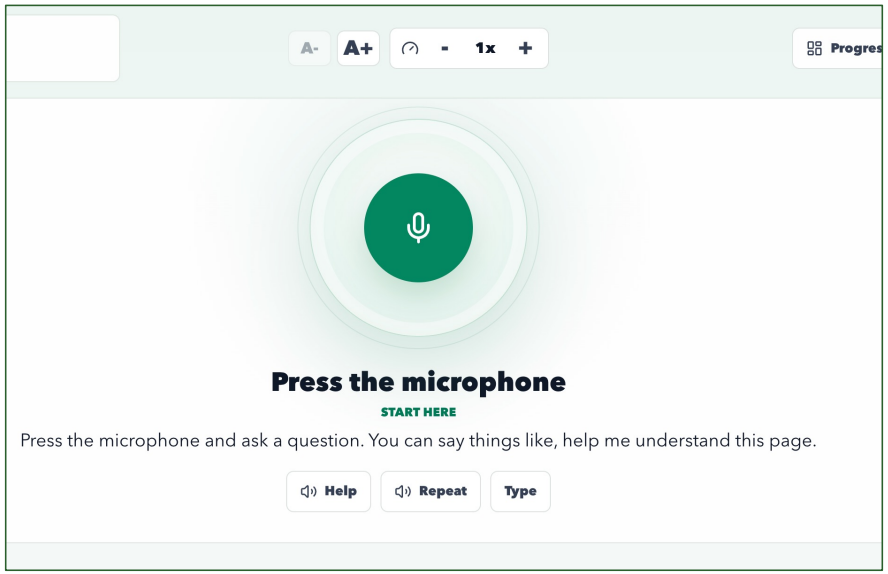
Two modes, one learning loop.

Phloem keeps the learner experience simple without hiding the system's depth.

VOICE LEARNING

For the learner

- Default first-run mode
- One-page capture flow
- Central microphone interaction
- Spoken tutor response with speech trace
- Minimal escape hatches: Help, Repeat, Type, Skip



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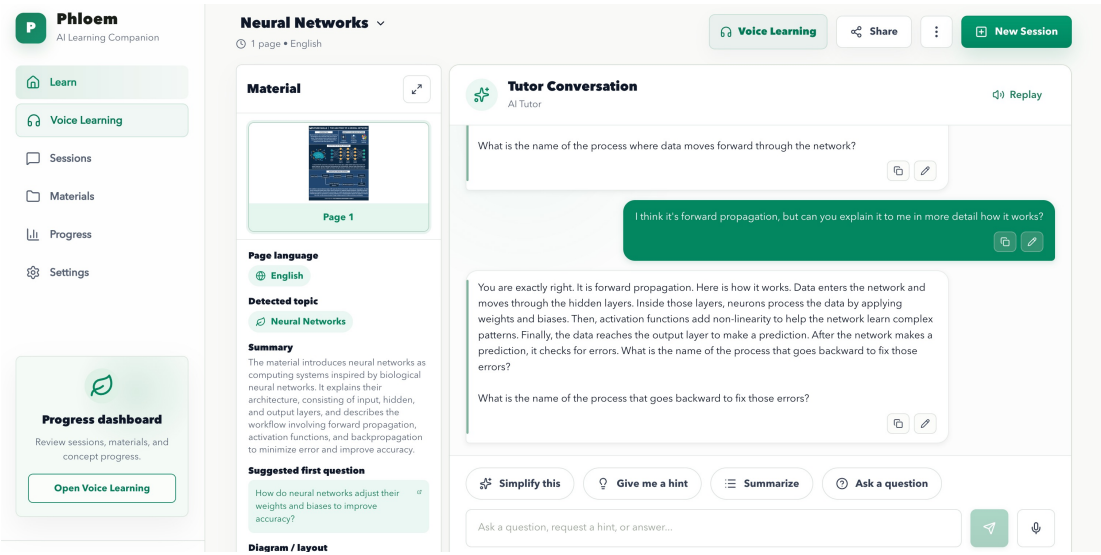


FULL DASHBOARD

For caregivers & power users

- Saved sessions & reloadable history
- Materials library with page previews
- Conversation panel with copy/edit controls
- Concept progress from retrieval checks
- Settings: language, voice, speed, TTS usage

Why this matters: *the same AI tutor supports accessibility-first learners and the adults or power users who need visibility, review, and control.*

REAL AI INTERACTION

Real AI calls across every step of the learning loop.

VISION

OpenAI GPT-4o

Reads images & rendered PDF pages

STT

GPT-4o-transcribe

Turns voice into text

TUTOR

Z.ai GLM-5.1

Structured Socratic responses

EVALUATOR

Z.ai GLM-5.1

Scores got-it / needs-practice / confused

TTS

Chirp 3 HD · OpenAI · Browser

Spoken output with fallbacks

TURN

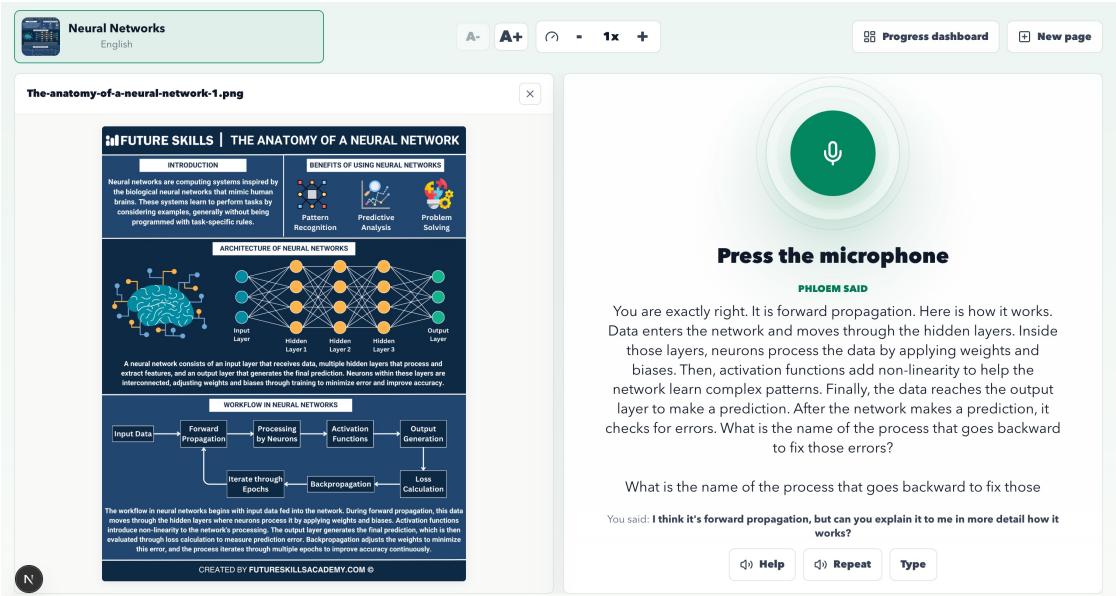
Silero VAD + Smart Turn

Detects when the learner stops talking

Not a scripted chatbot. *The AI reads source material, responds to learner turns, evaluates answers, and adapts memory.*

THE SOCRATIC TUTOR LOOP

Teach thinking before revealing answers.



01 Teach thinking before revealing final answers

02 Use short, spoken-friendly sentences

03 Ask exactly one focused follow-up question

04 Adapt to low, medium, or high understanding

05 Honor direct-answer mode only when allowed

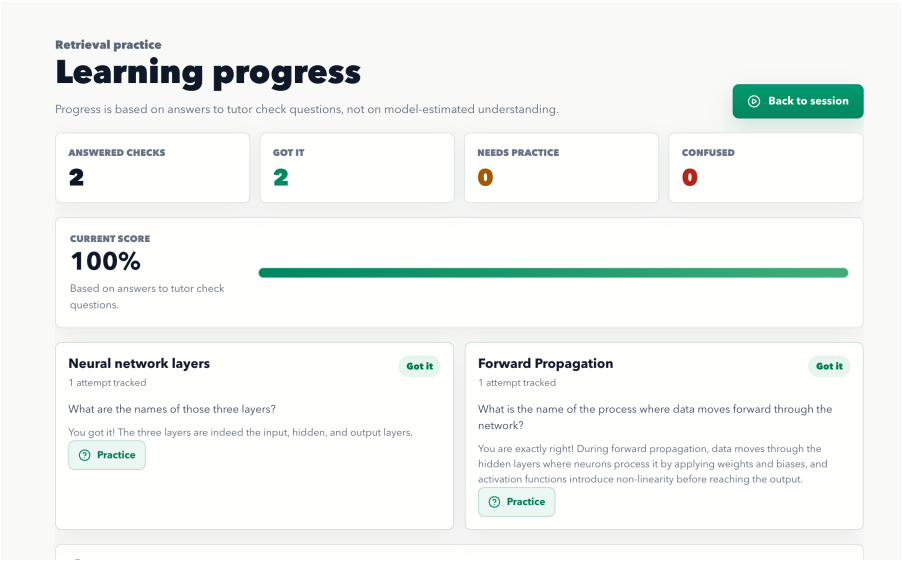
06 Treat uploaded material as untrusted context

MEASURING UNDERSTANDING

Evidence over vibes.

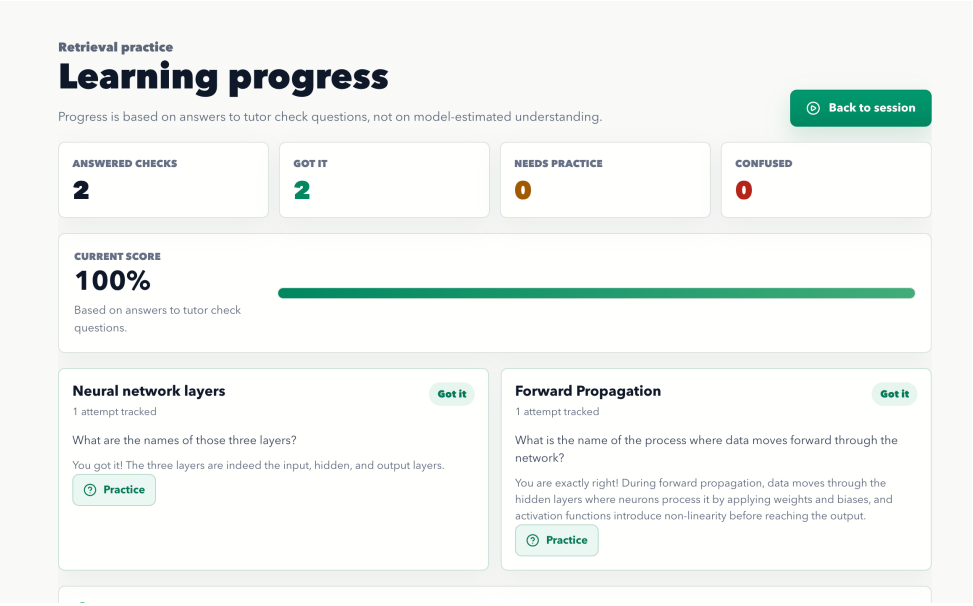
Phloem separates what the model thinks from what the learner actually demonstrates.

1. Tutor answer creates a check
2. Check appears after speech finishes
3. Learner answers by voice or text
4. Evaluator scores the answer
5. Concept mastery updates



PROGRESS DASHBOARD

A learning record — not just a chat transcript.



Answered
CHECKS COUNT

Got-it
TOTALS

Needs Practice
TOTALS

Confused
TOTALS

Score
CURRENT MASTERY

History
RECENT CHECKS

TECHNICAL EXECUTION

A functional Next.js + TypeScript app.

- **API Routes**
  Vision · tutor · evaluation · STT · TTS · sessions · materials · progress
- **Validation**
  Zod across request bodies and model outputs
- **Resilient parsing**
  Robust JSON handling for LLM responses
- **Document ingestion**
  PDF & DOCX extraction + rendered page images for OCR
- **Persistence**
  Supabase: sessions, materials, messages, memory, checks, mastery, TTS usage
- **Graceful fallbacks**
  When Supabase, Smart Turn, or paid TTS providers are unavailable

CLOSING

Phloem changes the default learning interface.

FROM

reading and prompting

→

TO

showing, listening, speaking, practicing

Phloem helps learners understand the page in front of them — one spoken step at a time.

Thank you.